Design Document of 25-26J-010

25-26J-010

AUTONOMOUS DATA FERRYING FROM A SELF-ORGANIZING, SELF-HEALING WIRELESS SENSOR NETWORK IN DISCONNECTED ZONES

STUDENT NAME	STUDENT ID
BANDARANAYAKE K. B. H. M. C. C.	IT22069122
BANDARA K. B. O. V.	IT22564986
WICKRAMA ARACHCHI D. R.	IT22360496
THAMASHA W. Y. M. G.	IT22088000

Supervisor: Ms. Dinithi Pandithage

11/01/2026

Table of Contents and Index

1 Introduction 3
1.1 Purpose 3
1.2 Scope 3
1.3 Definitions, Acronyms, and Abbreviations 4
1.4 Overview 6
2 Overall Descriptions 7
2.1 Product Perspective 7
2.1.1 System Interfaces 7
2.1.2 User Interfaces 9
2.1.3 Hardware Interfaces 9
2.1.4 Software Interfaces 11
2.1.5 Communication Interfaces 11
2.1.6 Memory Constraints 12
2.1.8 Site Adaptation Requirements 13
2.2 Product Functions 13
2.3 User Characteristics 14
2.4 Constraints 14
2.5 Assumptions and Dependencies 14
2.6 Apportioning of Requirements 15
3 Specific Requirements 16
3.1 External Interface Requirements 16
3.1.1 User Interfaces 16
3.1.2 Hardware Interfaces 17
3.1.3 Software Interfaces 18
3.1.4 Communication Interfaces 19
3.2 Architectural Design 19
3.2.1 High Level Architectural Design 19
3.2.2 Hardware and Software Requirements with Justification 21
3.2.3 Risk Mitigation Plan with Alternative Solution Identification 23
3.2.4 Cost Benefit Analysis for the Proposed Solution 24
3.3 Performance Requirements 24
3.4 Design Constraints 25

3.5 Software System Attributes 26
3.5.1 Reliability 26
3.5.2 Availability 26
3.5.3 Security 26
3.5.4 Maintainability 26
References 27

1 Introduction

1.1 Purpose

The purpose of this document is to provide a comprehensive technical blueprint for the development, implementation, and testing of the "Autonomous Data Ferrying" Wireless Sensor Network system. It is intended to serve as the definitive design reference for all project stakeholders, including developers, supervisors, and evaluators, ensuring a shared understanding of the system's architecture, components, and performance targets.

1.2 Scope

1. **Autonomous Multi-Sensor Nodes:** The hardware and software architecture of the sensor nodes, including their multi-modal sensor suite, power management engine, and local data storage mechanisms.
2. **Trust Enhanced Self-Organizing Clustering:** This component implements a fully decentralized, trust-enhanced self-organizing clustering mechanism, enabling sensor nodes to autonomously form clusters and elect a Cluster Head (CH) without centralized coordination or pre-configured roles.
3. **Fault-Tolerant Communication:** Energy-efficient, time-slotted ESP-NOW is used for reliable intra-cluster data transfer from Main Set nodes to the Cluster Head, with persistent local storage ensuring data survivability during failures, followed by acknowledgement-based data handover from the Cluster Head to the UAV to guarantee lossless delivery.
4. **Intelligent Unmanned Aerial Vehicle (UAV) Data Ferry:** The design of the UAV payload, including its mission manager system and data pipeline.
5. **Base Station Dashboard:** The basic design of the base station monitoring dashboard for visualizing UAV live telemetry data, a map with WSN nodes and their network status, UAV missions, and collected sensor data.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
WSN	Wireless Sensor Network - a network of spatially distributed sensor nodes that cooperatively monitor physical or environmental conditions.
MS	Main Set Node - a generic sensor node in the cluster (non-Cluster Head) that collects environmental data.
CH	Cluster Head - a sensor node elected to act as local aggregator and coordinator for a cluster of MSNs, responsible for intra-cluster communication and interfacing with the UAV.
UAV	Unmanned Aerial Vehicle - a drone used as a mobile data mule to periodically collect data from CHs and transport it to a base station.
BLE	Bluetooth Low Energy - a low-power wireless communication technology used in this project for node discovery.
ESP-NOW	A proprietary connectionless wireless protocol by Espressif that is used for fast, low-latency data transmission between ESP32 nodes (used for intra-cluster data exchange without Wi-Fi infrastructure).
SPIFFS	Serial Peripheral Interface Flash File System - a lightweight file system for on-chip flash memory on ESP32 microcontrollers, used for local data storage on nodes (especially at CH) to buffer sensor readings.
Coded PHY (BLE 5)	A physical layer option in Bluetooth 5 for long-range communication at the expense of data rate, used for extended-range BLE advertisements.
HMAC	Hash-Based Message Authentication Code - a cryptographic method for ensuring authenticity/integrity of messages. Used here to secure CH advertisement broadcasts.
RTC Memory	Real-Time Clock memory - a small persistent memory in microcontrollers that retains data during deep sleep. Used in nodes to store uptime counters across sleep cycles for fair CH election.
TDMA	Time Division Multiple Access - a channel access method where nodes transmit in assigned time slots to avoid collisions. Used in intra-cluster scheduling of sensor node transmissions.
EWMA	Exponentially Weighted Moving Average - a smoothing technique that reduces short-term noise by giving more weight to recent measurements while retaining historical influence.

RSSI	Received Signal Strength Indicator - a radio-level measurement (in dBm) representing the strength of a received wireless signal, used to estimate link quality.
PER	Packet Error Rate - the ratio of lost or corrupted packets to total transmitted packets, indicating communication reliability.
PDR	Packet Delivery Ratio - the proportion of successfully received packets.
NVS	Non-Volatile Storage - persistent flash memory used to store data such as uptime across device reboots.
Differential Entropy	An information-theoretic measure of uncertainty used to quantify how stable or unreliable a metric is over time.
Variance	Statistical measure representing how much a metric fluctuates between successive measurements.
Utility Function	A non-linear transformation that maps normalised metrics into dimensionless values reflecting their real-world importance.
Lyapunov Stability	Mathematical framework used to ensure that adaptive systems converge smoothly without oscillation or divergence.
Pareto Optimality	Condition where a solution cannot be improved in one metric without degrading at least one other metric.
Nash Bargaining Solution	A game-theoretic method that selects a fair compromise by maximizing balanced gains across multiple objectives.
Softmax Function	Normalization function that converts raw scores into relative confidence values that sum to one.
Disagreement Point	The minimum acceptable utility level for a metric in Nash bargaining, below which a candidate is considered unacceptable.
STELLAR	Stochastic Trust-Enhanced Lyapunov-stable Leader Allocation and Ranking - the proposed adaptive cluster-head election framework.
Huffman Coding	A lightweight, symbol-based lossless compression algorithm used to reduce the size of time-series sensor data.
miniz	A high-performance, dictionary-based lossless compression library (LZ-family) used for achieving higher compression ratios on embedded systems.
ROS 2	Robotic Operating system (version 2) - a distributed, middleware-based robotics framework designed for building scalable, real-time, and modular

	robotic applications, enabling reliable communication, hardware abstraction, and coordinated control across autonomous systems.
FCU (flight controller unit)	An open-source autopilot firmware providing reliable low-level flight control, navigation, and mission execution for multi-rotor UAVs, supporting autonomous operations, sensor fusion, and failsafe handling
Ard Pilot SITL (Software-In-The-Loop)	A software-based simulation environment that runs the ArduPilot Copter firmware in a virtual setup, enabling safe testing, debugging, and validation of flight control logic, mission planning, and ROS 2-based integrations without requiring physical UAV hardware.

1.4 Overview

The primary goal of this system is to enable long-term, autonomous environmental monitoring in disconnected zones where conventional communication infrastructure is absent. The system accomplishes this through four steps. First, low-power MS nodes collect environmental data and autonomously self-organize into clusters. Second, a secure and robust algorithm elects a CH within each cluster to aggregate and persistently store this data. Within each cluster, MS-to-CH communication is performed using time-sliced, collision-free ESP-NOW transmissions, where MS nodes wake only during assigned slots to transmit buffered data, minimizing energy consumption and ensuring reliable data aggregation at the CH. Finally, an intelligent UAV periodically flies missions to retrieve the buffered data, perform network health audits. This data is then offloaded to a base station for analysis and visualization. The primary users of this system are environmental scientists, conservationists, disaster response agencies, and agricultural professionals who require reliable field data from remote locations. The following sections will provide a more detailed description of the system's architecture and technical requirements.

2 Overall Descriptions

This section provides a high-level, holistic view of the system, describing its context, core functions, and operational constraints. It is intended to establish a stable and comprehensive understanding of the product's purpose and its place within the broader field of environmental monitoring. This overview provides the necessary background before diving into the specific technical requirements detailed in subsequent sections.

2.1 Product Perspective

The proposed system represents a significant evolution in environmental monitoring technology. Conventional WSNs typically rely on static sinks or multi-hop forwarding to transmit data, which rapidly depletes node batteries and limits operational range in infrastructure-less areas. This project's UAV-assisted "data mule" architecture directly addresses these limitations. By physically transporting data, the UAV decouples the sensing network from the need for a continuous communication backbone. This approach dramatically extends network lifetime by allowing sensor nodes to use low-power, short-range communication protocols and remain in deep sleep for extended periods. It also enhances scalability and operational range, enabling deployments in vast, inaccessible, or hazardous terrains.

2.1.1 System Interfaces

1. **Real-Time Operating System (RTOS) Services:** The sensor nodes (ESP32-S3) utilize the ESP-IDF framework, which provides a FreeRTOS interface.
    - **Task Scheduling:** The software interfaces with FreeRTOS to manage dedicated tasks for periodic scoring, neighbor scanning, and BLE status announcements.
    - **Asynchronous Processing:** On the UAV side (Raspberry Pi), the system uses multi-threaded or asynchronous Python tasks to handle parallel processing for health auditing and machine learning inference.
2. **System Timers:** Precise timing is critical for synchronizing the WSN.

- **RTC (Real-Time Clock):** The software interfaces with the system's RTC to provide uptime persistence across deep sleep cycles and to timestamp environmental data.
- **TDMA Scheduling:** A Time-Sliced scheduling model is implemented via system timers to ensure collision-free data transmission; the CH assigns unique slots to MS nodes, requiring precise hardware clock alignment.
- **Watchdog Timers:** The system includes an auto-reboot function triggered by hardware watchdog services if the software crashes.

3. **Power Management Services:** The Power Management Engine (PME) is the software's primary interface with the hardware's energy states.

- **Operational Modes:** The software interfaces with system power services to transition nodes between Normal, Passive (Power-Save), and Critical (Deep-Sleep) modes based on battery thresholds.
- **Deep Sleep Control:** The system invokes the ESP32's deep sleep to drop current consumption to the order of tens of microamperes when inactive.
- **ADC Monitoring:** The software interacts with the system's Analog-to-Digital Converter (ADC) to read data from the INA219 current sensor, allowing it to calculate the remaining energy and solar charging rates.

4. **File System Services:** Local storage is essential because the nodes operate in "disconnected zones".

- **SPIFFS (Serial Peripheral Interface Flash File System):** The software uses the SPIFFS interface on the ESP32 for non-volatile, persistent storage of sensor data.
- **Structured Storage:** The software interacts with the file system via the cJSON library to store readings in a structured chunk format with CRC32 integrity checks.
- **UAV Storage:** The UAV's companion computer manages local data logging and a temporary Object Store for data retrieved from the field.

5. **Network Stack Services:** The system manages a hybrid multi-layered communication stack.

- **ESP-NOW:** The software interfaces with this connectionless, low-power peer-to-peer protocol for rapid intra-cluster data exchanges.

- **BLE 5.1 Coded PHY:** Used for long-range, low-power control signaling.
- **Wi-Fi:** Once xy-mk-5v module connection is established, the software triggers the Wi-Fi stack to function as an WIFI-ACCESSPOINT for bulk data offloading to http server.
- **MAVLink Protocol:** This acts as the software interface between the high-level logic on the Raspberry Pi and the low-level Pixhawk flight controller.

2.1.2 User Interfaces

The primary user interface is a web-based dashboard located at the base station. This dashboard serves as the central command and control hub, providing operators with a real-time, map-based visualization of the WSN. Key features include:

- A map displaying the discovered locations of clusters and their CHs.
- Node health status indicators (e.g., Normal, Power-Save, Critical).
- Visualization of the UAV's current flight path and planned routes.
- Access to historical and real-time sensor data logs.

2.1.3 Hardware Interfaces

1. **UAV Platform (F450 Drone Kit & Flight Control):** The software relies on flight controllers and specialized RF modules for autonomous operation and node triggering:

- **Pixhawk 2.4.8 FCU**: Runs Copter-4.6.3 and interfaces with the companion computer via MAVLink for autonomous mission execution and telemetry.
- **XY-MK-5V 433 MHz RF Transmitter**: A critical interface used by the UAV software to send a short RF pulse that triggers the sensor nodes' wake-up pins.
- **Realtek USB-Wi-Fi Adapter**: Attached to the companion computer to establish a high-throughput Wi-Fi link for bulk data retrieval from Cluster Heads (CH).
- **F450 Kit Components:** Includes brushless motors, 40A ESCs, and propellers which are controlled by the flight stack to manage altitude and hovering stability during data collection.

2. **UAV Companion Computer (Raspberry Pi 4 Model B):** The Raspberry Pi 4 (running Ubuntu 22.04.5 Server) serves as the "brain" for high-level software tasks:

- **Onboard Processing:** machine learning models for energy prediction, and the Health Auditor.
- **Storage:** Local storage for buffered JSON data retrieved from the field before relaying it to the base station.

3. Sensor Nodes (ESP32-S3-N16R8): The nodes are specialized for environmental sensing and ultra-low-power persistence:

- **ESP32-S3-N16R8 Microcontroller:** Manages the Power Management Engine (PME), local SPIFFS storage, dual-plane communication (BLE and ESP-NOW), and on-node compression.
- **Multi-Sensor Suite:**
    - ENS160 + AHT21: For air quality and volatile organic compound (VOC) levels.
    - INMP441 Digital MEMS Microphone: Connected via I2S for bio-acoustic monitoring.
    - QMC5883L Magnetometer: For magnetic field intensity.
    - BME280: For pressure, temperature, and humidity.
- **Special Trigger & Power Hardware:**
    - XY-MK-5V 433 MHz Receiver: Wired directly to the ESP32 wake-up pin to transition the node from deep sleep to active mode.
    - From CH to UAV data is transport using HTTP/HTTPS protocol to a local server in the onboard-companion computer (at this stage the data is still compressed).
    - INA219 Current Sensor: Provides voltage and current data to the software to calculate remaining battery life and solar charging efficacy.
    - Solar Power Unit: 6V 1W solar panel and CN3065 charger to enable long-term energy autonomy.

2.1.4 Software Interfaces

- **MAVLink:** This protocol facilitates communication between the Raspberry Pi 4 companion computer and the Pixhawk flight controller, allowing the Raspberry Pi to issue high-level commands and receive telemetry data.
- **UAV-Assisted Data Ferrying Pipeline:** A cluster-based wireless sensor network operates in a disconnected environment where sensor nodes forward readings to an energy-efficiently elected Cluster Head (CH). A UAV periodically visits these CHs to collect the aggregated data and transports it to a ground base station, where the data is offloaded and visualized through a web application for monitoring and analysis.
- **Ardupilot Firmware (Copter-4.6.3):** UAV path plaining is handled from this software interface (connected to raspberry pie 4 model B via onboard GPIO pins and the onboard companion computer is directly powered over that link too).
- **ROS2:** Base of the **SKY-MULE** purposed solution. Generate mission logic for given GPS location list (order matter, no auto-sorting functionality). A single ROS2 node for data onboarding and data bucked and offloading.
- **Operations and Control Dashboard:** A unified interface that enables monitoring, configuration, lifecycle management (**WSN-restart**, **WSN-hold** (waiting for full battery), **WSN-terminate**), and mission control of a clustered wireless sensor network and the associated UAV data ferry, with real-time geospatial visualization.

2.1.5 Communication Interfaces

- **ESP-NOW:** Used for energy-efficient, low-latency, connectionless data transfer from MS nodes to the CH within a cluster. Its minimal overhead is ideal for nodes that need to wake, transmit, and return to sleep quickly.
- **Bluetooth Low Energy (BLE):** Serves multiple critical roles:
    - **Proximity Detection:** Low-power beaconing allows the CH to detect the UAV's arrival.
    - **Secure Announcements:** HMAC-signed extended advertisements are used for secure CH announcements, preventing spoofing during cluster formation.
    - **HMAC** verification results contribute to the trust metric for STELLAR algorithm.

- **Reliability Inputs from Communication**: Packet reception success and sequence gaps provide packet error rate (PER) and packet delivery ratio (PDR), feeding trust and link-quality computation.

- **Wi-Fi:** Provides a reliable, acknowledged channel for the bulk transfer of buffered data from the CH to the UAV once proximity is established.
- **Telemetry Link:** A long-range radio link enables real-time communication between the UAV in flight and the base station dashboard, relaying mission status and critical alerts.

2.1.6 Memory Constraints

To ensure data integrity between infrequent UAV visits and to provide resilience against power cycles or node failures, sensor nodes face memory constraints. This is addressed by utilizing the onboard SPIFFS flash memory on the ESP32-S3 microcontrollers for persistent data buffering. Data is stored in a structured JSON format at the Cluster Head (CH) until it is successfully offloaded to the UAV and acknowledged. To optimize the utilization of the limited SPIFFS flash memory, the system supports on-device data compression, reducing the storage footprint of verbose JSON records and enabling larger volumes of environmental data to be buffered between UAV visits. In the event of a CH failure or re-election, the buffered data is transferred to thte newly elected CH, allowing data collection to continue without loss or reinitialization.

2.1.7 Operations

- **Normal Operations:** These include the routine, periodic tasks of the WSN, such as environmental sensing by MS nodes, data buffering, and time-sliced, scheduled MS to CH communication, data offloads and viewed.
- **Special Operations:** These are event-driven or mission-specific tasks, such as the dynamic re-election of a CH in response to a failure, the execution of specific UAV mission modes (waypoint, hover-networ_audit), and aerial health auditing sweeps to detect silent or failing nodes.

2.1.8 Site Adaptation Requirements

The system is purpose-built for deployment in remote, disconnected zones where no permanent communication infrastructure, such as cellular networks or internet access, is available. Target environments include wildlife preserves, agricultural fields, disaster areas, and other hard-to-reach locations requiring long-term, unattended monitoring.

2.2 Product Functions

- **Autonomous Multi-Modal Sensing:** Deploys sensor nodes capable of collecting a diverse range of environmental data (air quality, acoustic, magnetic, atmospheric).
- **Self-Organizing Cluster Formation:** Enables nodes to autonomously discover neighbors and form logical clusters without manual configuration through the STELLAR algorithm.
- **STELLAR Functions:** Neighbor discovery and neighbor table maintenance using BLE. Trust computation using security correctness (HMAC success), communication reliability (PDR), and reputation. Link quality estimation using RSSI EWMA combined with packet reliability. Uncertainty modeling via EWMA variance estimation. Confidence modeling via differential entropy and softmax normalization. Adaptive metric weight updates using Lyapunov-stable convergence constraints. Cluster Head election using Pareto frontier extraction and Nash bargaining selection. Self-healing re-election under battery/trust/link degradation or CH conflicts.
- **Secure and Resilient Data Aggregation:** Implements a robust, multi-parameter algorithm for electing a CH, which securely aggregates and persistently stores data from its cluster members.
- **Energy-Efficient Intra-Cluster Communication:** Uses time-sliced ESP-NOW communication for MS to CH data transfer, allowing nodes to transmit only in assigned slots to reduce collisions and conserve energy.
- **Intelligent UAV-Based Data Mulling:** Utilizes a UAV to perform GPS-free localization, execute energy-aware dynamic routing, and reliably ferry collected data from CHs to a base station.

- **Centralized Network Health Monitoring:** Provides a base station dashboard for operators to visualize the WSN's status, monitor node health, and receive alerts about potential failures.
- **Efficient Data Payload Optimization:** Implements a firmware-level compression strategy to minimize the data volume requiring transmission. By reducing the raw payload size, the system maximizes the data yield that can be ferried during short "data mule contact" windows.

2.3 User Characteristics

The typical users of this system are professionals such as environmental scientists, ecologists, agricultural managers, or disaster management personnel. These users require reliable and continuous field data to support their research or operations but are not necessarily experts in embedded systems, network engineering, or UAV piloting. The system is designed to be highly autonomous to meet their needs with minimal technical intervention.

2.4 Constraints

- **Lack of Permanent Infrastructure:** The system must operate entirely offline, without reliance on cellular, Wi-Fi, or internet connectivity in the deployment zone.
- **Finite Power Resources:** Sensor nodes have limited battery life, which is mitigated by solar recharging but still necessitates highly energy-efficient operation. The UAV's flight endurance is also finite, requiring optimized mission planning.
- **Constrained Computational Resources:** The ESP32 microcontrollers on the sensor nodes have limited processing power and memory, constraining the complexity of on-node algorithms.

2.5 Assumptions and Dependencies

- **Periodic Solar Availability:** The long-term operation of sensor nodes assumes sufficient periodic sunlight to enable the solar units to recharge the batteries.
- **UAV Navigability:** The design depends on the UAV's ability to safely and effectively navigate the target terrain to come within communication range of the CHs.

2.6 Apportioning of Requirements

The system's requirements will be implemented in a phased approach. This will begin with the prototyping and testing of individual components, such as the multi-sensor node and the UAV. Subsequently, subsystem integration will be performed to validate intra-cluster communication and UAV-CH data exchange. The final phase will involve full end-to-end field testing to evaluate the performance of the complete, integrated system. This structured progression ensures that each component is validated before contributing to the complexity of the final system, which is detailed further in the following section.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The base station dashboard will be a web-based application designed for intuitive operation and comprehensive network oversight. It will be built with a React frontend for a responsive and interactive user experience and a Spring Boot microservice backend for robust data processing. A PostgreSQL database will be used for persistent storage of all collected sensor data, network health logs, and mission parameters.

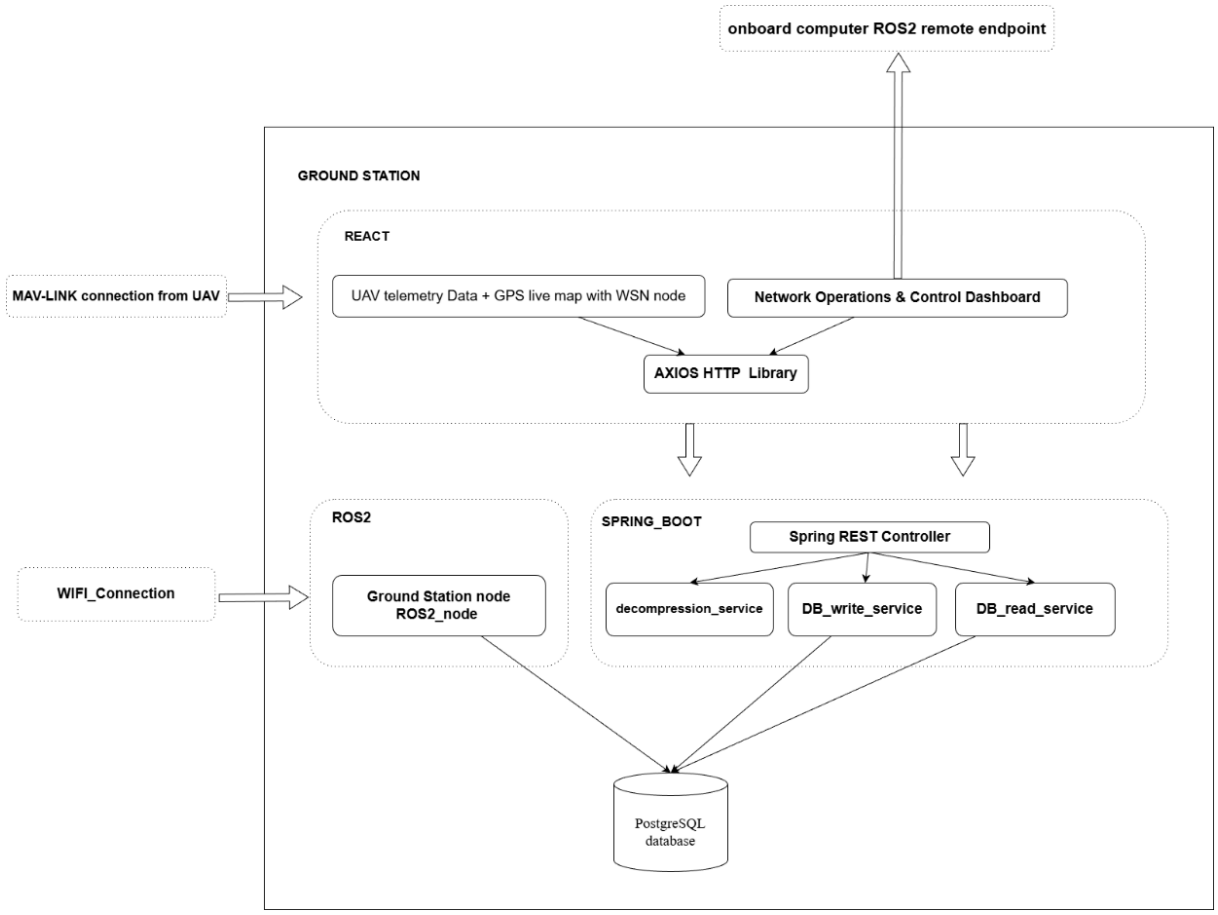


Figure 3.1: Base Station Architecture

The dashboard's primary function is to display a live, map-based visualization of WSN. This interface will enable operators to:

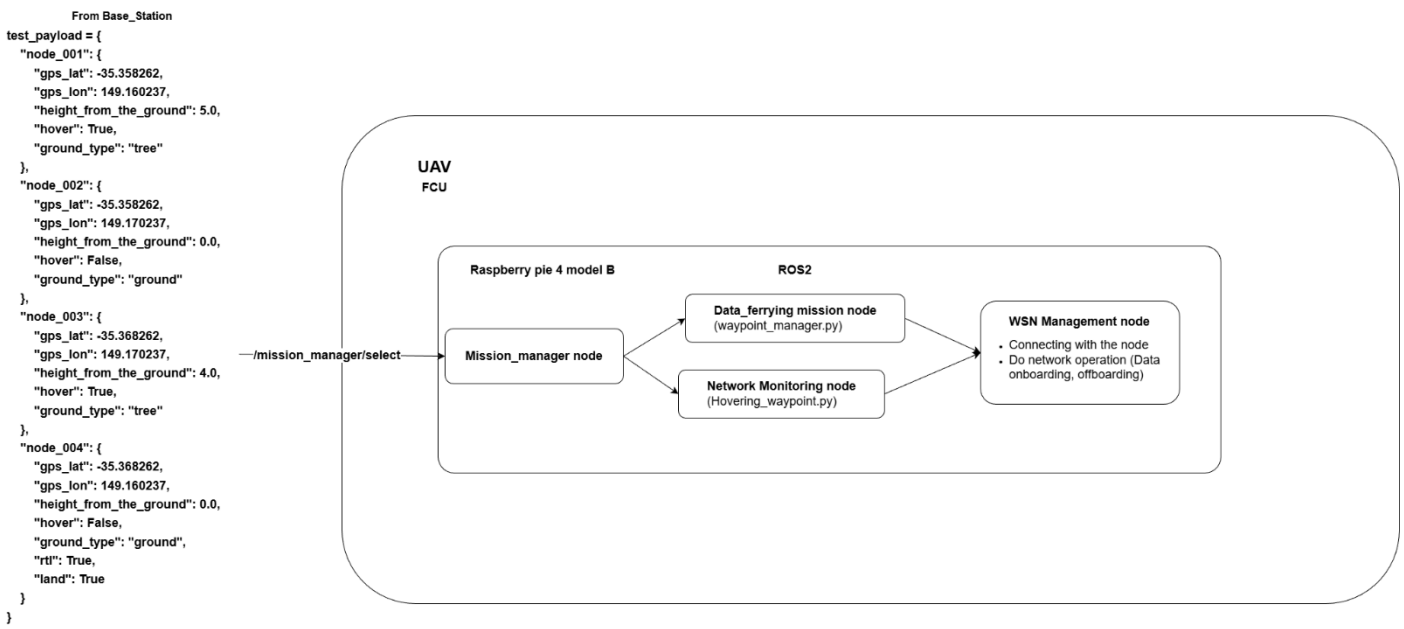
- View the geographic locations of all active CHs.
- Monitor prioritized UAV routes calculated by the energy-aware routing engine.
- Receive and acknowledge flagged health alerts (e.g., "CH Battery Critical"), enabling proactive operator intervention and maintenance planning.

3.1.2 Hardware Interfaces

- **UAV Platform:**
    - **Frame:** F450 Quadcopter frame, providing a stable and robust platform with carbon fiber landing gear.
    - **Flight Controller:** Pixhawk 2.4.8, serving as the primary autopilot.
    - **Companion Computer:** (Raspberry Pi 4 Model B) A Linux-based onboard computing platform responsible for high-level autonomy, mission coordination, and data handling, running ROS 2 Humble to manage inter-node communication, sensor data processing, UAV-WSN interaction, and payload offloading to the ground station**.**
- **Sensor Nodes (MS/CH):**
    - **Microcontroller:** ESP32-S3, providing dual-core processing, integrated Wi-Fi and BLE, and low-power modes.
    - **Sensor Suite:** Integrated sensors for multi-modal environmental monitoring, including ENS160 + AHT21 (air quality), INMP441 (acoustic), QMC5883L (geomagnetic), and BME280 (atmospheric).
    - **Power System:** A solar charging unit based on the CN3065 module connected to a Li-ion battery.
    - **Wake-Up Receiver:** A 433 MHz RF receiver (XY-MK-5V) is integrated to enable remote activation from deep sleep. The UAV initiates contact with a long-range 433 MHz RF signal, waking the CH. Once awake, the CH begins broadcasting low-power BLE beacons for localization.

3.1.3 Software Interfaces

- **Firmware Development:** ESP-IDF (Espressif IoT Development Framework) and the Arduino IDE are used for developing the firmware for ESP32-based sensor nodes.
- **UAV Flight Control:** (Arduino Copter v4.6.3) An open-source autopilot firmware providing reliable low-level flight control, navigation, and mission execution for multirotor UAVs, supporting autonomous operations, sensor fusion, and failsafe handling.
- **Onboard Computing (UAV):** Raspberry Pi 4 Model B with Ubuntu Server 22.04.5 for ROS 2 Humble. Python scripts run on the ROS2 nodes to manage the mission. These scripts are leverage libraries such as FastAPI and rclpy for http servers for data buckets and ros2 functionalities, and dedicated libraries for MAVLink and WI-FI communication.



- **Simulation Environment:** ROS2 + Gazebo + Ardupilot SITL are utilized for simulating UAV flight dynamics, testing routing algorithms, and validating mission logic in a controlled virtual environment before field deployment.

3.1.4 Communication Interfaces

- **BLE 5.1 Coded PHY:** This long-range Bluetooth mode is used for the initial discovery of CHs by the UAV. Its enhanced sensitivity provides a robust link for discovery even in Non-Line-of-Sight (NLoS) conditions, overcoming the limitations of standard BLE or RSSI-based methods.
- **ESP-NOW:** This connectionless protocol is used for intra-cluster communication. MS nodes transmit data in short, efficient bursts to the CH without the overhead of a full Wi-Fi association, conserving significant energy.

3.2 Architectural Design

3.2.1 High Level Architectural Design

The system architecture is composed of three interconnected, yet functionally distinct, components: the Self-Organizing WSN, the Intelligent UAV Data Ferry, and the Base Station.

1. **Self-Organizing WSN:** The foundation of the system is a network of autonomous sensor nodes. The lifecycle of a node begins with deployment, followed by a neighbor discovery phase using BLE broadcasts. Nodes then participate in a secure CH election process. The election algorithm is multi-parametric, evaluating candidates based on residual battery level, operational uptime, a computed trust score, and link quality. This ensures the most capable node is chosen as the CH. Once a cluster is formed, MS nodes enter a time-sliced operational state, waking only during their assigned slots to transmit sensor data to the CH via the energy-efficient ESP-NOW protocol. All received data is persistently stored in the CH's SPIFFS flash memory to survive reboots or power failures.
2. **On-Device Compression Benchmarking:** The firmware includes a dedicated benchmarking task to evaluate and compare the performance of Huffman coding versus miniz. This task profiles each algorithm based on:
    a. **Compression Ratio:** Percentage reduction of the JSON-formatted sensor data.

b. **Execution Time:** CPU cycles required for compression, ensuring it does not interfere with real-time sensing.
c. **Energy Overhead:** The trade-off between the energy saved during shortened transmission and the energy consumed during CPU-intensive compression.

3. **Intelligent UAV Data Ferry:** The UAV acts as an active, intelligent agent, not just a passive data mule. Its mission workflow is executed in three distinct phases:
    a. **Phase 1: Dynamic Routing:** The UAV utilizes an onboard machine for management of waypoint missions. Based on its current mission mode (waypoint missions or hover missions), it prioritizes its route to visit nodes that are most in need of data collection, for both network monitoring and network management (TERMINATION, RESET).
    b. **Phase 2: Health Auditing & Data Mulling:** As it approaches a CH, the UAV probes for silent or failing nodes in the vicinity, flagging health anomalies for the base station. It then establishes a Wi-Fi connection with healthy CH and collects all buffered data (formatted as JSON) via the HTTP/HTTPs protocol before proceeding to its next target.
4. **Base Station:** The Base Station is the central monitoring and data aggregation hub. After completing its mission, the UAV offloads all collected sensor data and decompresses the data, offloads health audit logs to the base station server. This data is processed and fed into the operator dashboard for real-time visualization, historical analysis, and the generation of alerts, closing the loop on the data collection cycle.
5. **Trust-Enhanced Self-Organizing Clustering (STELLAR):** After deployment, sensor nodes autonomously discover neighboring nodes through low-power BLE signaling and form logical clusters without centralized control. Cluster Head (CH) election is performed using a secure, multi-parameter algorithm that evaluates each candidate based on battery level, persistent uptime, computed trust score, and link quality. Unlike static-weight approaches, the proposed STELLAR framework incorporates noise awareness by estimating metric variance and deriving entropy-based confidence values. These confidences drive Lyapunov-stable adaptive weight updates, ensuring that leadership decisions converge smoothly without oscillation in

stochastic wireless conditions. The elected CH becomes the cluster coordinator and the designated interface node for UAV-based data ferrying.

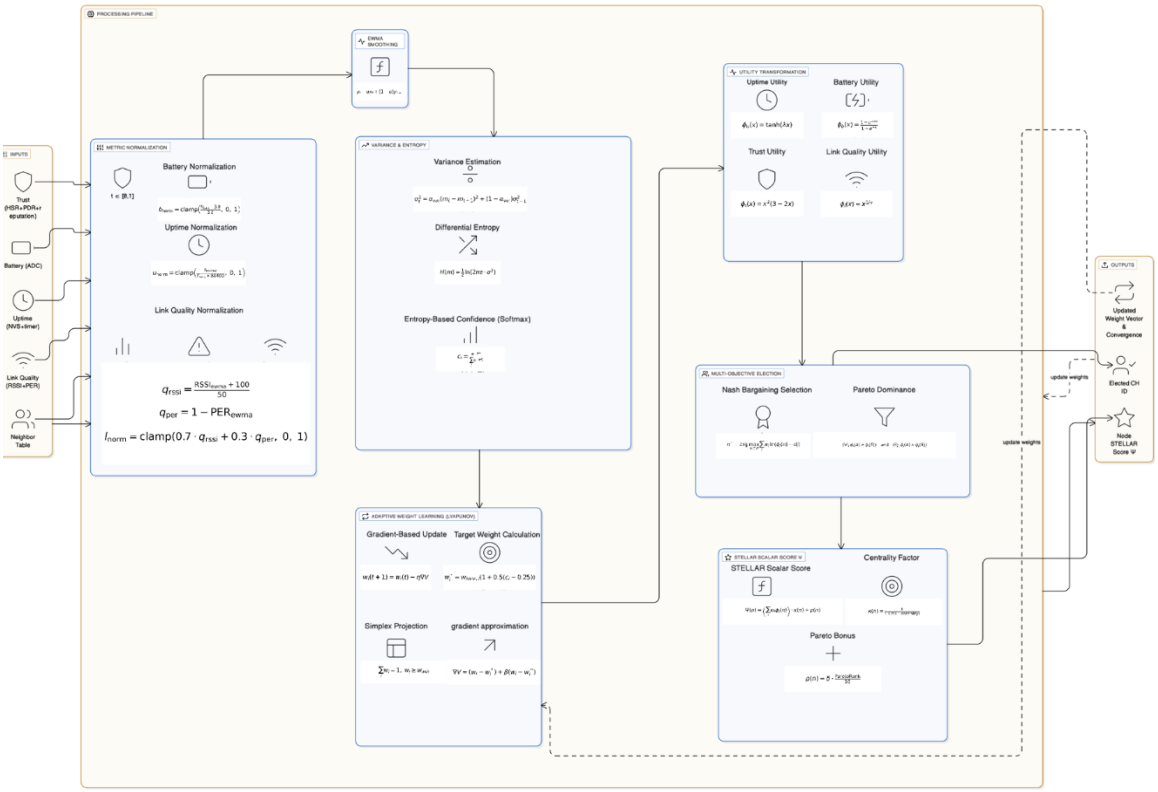


Figure 3.2: CH Detection Algorithm High-Level Design

This algorithm reduces leadership instability under stochastic wireless behavior and prevents noisy metrics from dominating CH selection, supporting energy efficiency and operational robustness

3.2.2 Hardware and Software Requirements with Justification

Hardware Requirements

Component	**Justification**
F450 Quadcopter Frame	Provides a stable, durable, and cost-effective airframe with sufficient payload capacity for the required electronics.

Pixhawk 2.4.8	A robust, open-source flight controller with extensive support for autonomous missions and companion computer integration.
Raspberry Pi 4 model b	A powerful and versatile single-board computer, ideal for running the high-level mission logic, ML models, and communication protocols on the UAV.
ESP32-S3 Microcontroller	Chosen for its dual-core processing power, integrated Wi-Fi and BLE 5 capabilities, and extensive I/O.
ENS160, INMP441, etc.	A suite of digital sensors selected to provide a comprehensive, multi-modal view of the environment (air, sound, etc.) with low power consumption.
CN3065 Solar Charge Module	A compact and efficient module specifically designed for single-cell Li-ion battery charging from a solar panel, enabling energy autonomy for the nodes.

Software and Protocol Requirements

Software/Protocol	**Justification**
ESP-IDF/FreeRTOS	The official development framework for ESP32, providing robust libraries and a real-time OS (FreeRTOS) necessary for managing complex, concurrent tasks on the sensor nodes.
ArduPilot Firmware	Industry-standard open-source autopilot firmware that provides the foundation for safe and reliable autonomous UAV navigation.
MAVLink Protocol	A lightweight messaging protocol for communicating with drones, essential for the Raspberry Pi to command the Pixhawk and receive telemetry.

ROS & Gazebo	A powerful simulation framework that allows for rapid prototyping, testing, and validation of UAV algorithms (e.g., routing, localization) before field deployment.
Python	The language of choice for the Raspberry Pi due to its extensive libraries for scientific computing and networking.
ESP-NOW	Chosen for intra-cluster communication due to its extremely low latency and connectionless nature, allowing MS nodes to transmit data and return to sleep with minimal energy expenditure.

3.2.3 Risk Mitigation Plan with Alternative Solution Identification

- **Risk: CH Node Failure**
    - **Mitigation:** Data from MS nodes is buffered persistently on the CH's SPIFFS. Furthermore, a robust and periodic CH re-election algorithm ensures that if a CH fails, a new one is quickly elected from the healthy nodes in the cluster, preventing cluster collapse and data loss. MS nodes retain their data until they confirm it has been received by a CH.
- **Risk: Data Loss during Transfer**
    - **Mitigation:** The CH-to-UAV data transfer uses the HTTP protocol configured with a Quality of Service (QoS) level that requires acknowledgement. The CH will only clear its local data buffer after receiving explicit confirmation from the UAV's HTTP reply header that the data has been successfully received.
- **Risk: Premature Node Power Depletion**
    - **Mitigation:** This risk is addressed by a two-pronged strategy. First, the integrated solar recharging unit provides a renewable energy source. Second, the firmware includes a Power Management Engine (PME) with three operational modes (Normal, Passive, Critical) that automatically throttles node activity based on the remaining battery level to ensure survival.

- **Risk: Inefficient UAV Routing**
    - **Mitigation:** The UAV follows pre-defined GPS waypoint-based missions generated offline, relying solely on the Pixhawk 2.4.8 flight controller and NEO-M8N GPS module for navigation. To reduce routing inefficiencies, waypoint sequencing is optimized at the mission-planning stage. Obstacle avoidance is not performed onboard; therefore, missions are restricted to open or previously surveyed environments

3.2.4 Cost Benefit Analysis for the Proposed Solution

The estimated total budget for the prototype, including the hobby-type UAV and multiple sensor nodes, is approximately 245,030.00 LKR. The UAV used in this prototype has a limited flight time of 10-15 minutes per mission, and the sensor nodes have a maximum range of around 100-200 meters line-of-sight. The UAV used in this prototype has a limited flight time of 10-15 minutes per mission, and the sensor nodes have a maximum range of around 100-200 meters (LoS). Standard SMA antennas sourced from Amazon are used for this test version, rather than custom-tailored solutions, which limits the operational range and performance.

While this is an initial prototype, it serves as proof of concept for UAV-assisted data collection from clustered WSNs in small, controlled areas. The prototype allows testing of mission planning, data aggregation, and UAV -WSN interaction, providing valuable insights for future improvements, such as longer flight times, enhanced communication range, and custom antenna designs. This realistic assessment ensures that the system's capabilities, limitations, and next steps are clearly understood.

3.3 Performance Requirements

- **Data Delivery Reliability:** Achieve a success rate of >99% for acknowledged CH-to-UAV data transfers.
- **Data Offload Latency:** The use of on-device compression is a primary mechanism to guarantee that the CH-to-UAV transfer window (< 5 seconds) is maintained regardless of the accumulated sensor data volume. This directly supports the goal of reducing the UAV's hovering time and conserving its finite battery endurance.

- **UAV Endurance:** The f450 platform is expected to provide ~10-15 minutes of operational flight time per mission.
- **Fault Detection Accuracy:** The UAV's aerial health audit must achieve >95% accuracy in correctly identifying silent or failing nodes.
- **Election Completion Time:** < 5 s within election window under up to N candidates.
- **Leader Stability:** Avoid frequent CH switching; target < 1 re-election per window under stable conditions.
- **Trust Responsiveness:** Detectable trust degradation within $\leq 3\tau$ of EWMA time constant.
- **Metric Update Overhead:** < 5% CPU utilization on ESP32 under normal conditions (to preserve sensing and communication).
- **Security Integrity:** CH announcements must be verifiable; invalid HMAC events must reduce HSR rapidly.

3.4 Design Constraints

- **Computational Limits:** The system must operate within the limited computational power and memory of the ESP32-S3 microcontrollers, necessitating lightweight algorithms for on-node processing. Compression algorithms are restricted to lightweight implementations (Huffman and miniz) that can operate within the limited RAM and dual-core processing limits of the ESP32-S3 without causing system instability or triggering watchdog timeouts.
- **Physical Platform Limits:** The design is constrained by the physical payload capacity and finite flight time of the S500 drone platform.
- **RF Environment:** Communication protocols must be robust enough to function reliably in challenging radio frequency (RF) environments, including both Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) scenarios with potential for multipath fading and interference.

3.5 Software System Attributes

3.5.1 Reliability

System reliability is a cornerstone of the design, achieved through a combination of fault-tolerant mechanisms. Persistent storage on SPIFFS ensures data survives node reboots. Acknowledged data transfer protocols (HTTP with ACK headers) prevent data loss during offload. The secure and resilient CH election algorithm ensures network leadership is always maintained, and the self-healing Power Management Engine (PME) allows nodes to gracefully handle low-energy conditions.

3.5.2 Availability

Despite the disconnected nature of the deployment, the system is designed to maintain high data availability. The store-and-forward mechanism, where nodes buffer data locally until a collection opportunity arises, coupled with periodic UAV ferrying, ensures that all collected data is consistently and eventually delivered to the base station. This architecture guarantees that no data is permanently lost due to temporary network partitions or node unavailability.

3.5.3 Security

STELLAR incorporates cryptographic integrity through HMAC-based authentication of leadership-related messages. Trust scoring integrates security correctness (HSR) with reliability indicators (PDR), preventing nodes that exhibit suspicious behavior or message tampering from being elected as Cluster Heads. This reduces CH spoofing and improves resilience against adversarial leadership manipulation in disconnected environments.

3.5.4 Maintainability

The system is designed for high maintainability through modular architecture. The sensor node firmware, UAV payload software, and base station dashboard are developed as independent, decoupled components. This separation of concerns facilitates targeted debugging, testing, and upgrades without affecting the entire system. The architecture also allows for the potential implementation of Over-the-Air (OTA) firmware updates, enabling remote maintenance and feature enhancements.

References

[1] O. Tsilomitrou and A. Tzes, “Mobile Data-Mule Optimal Path Planning for Wireless Sensor Networks,” Applied Sciences, vol. 12, no. 1, p. 247, Dec. 2021.

[2] W. Heinzelman, A. Chandrakasan, and H. Balakrishnan, “Energy-efficient communication protocol for wireless microsensor networks,” in Proc. 33rd Annu. Hawaii Int. Conf. Syst. Sci., 2000, pp. 1-10.

[3] S. Pudlewski, T. Melodia, and A. Prasanna, “Compressed sensing for real-time energy-efficient wireless multimedia sensor networking,” IEEE Commun. Mag., vol. 50, no. 6, pp. 40-49, Jun. 2012.

[4] M. Younis and K. Akkaya, “Strategies and techniques for node clustering in wireless sensor networks: A survey,” Ad Hoc Netw., vol. 6, no. 4, pp. 621-648, Jun. 2008.

[5] I. Demirkol, C. Ersoy, and F. Alagöz, “MAC protocols for wireless sensor networks: A survey,” IEEE Commun. Mag., vol. 44, no. 4, pp. 115-121, Apr. 2006.

[6] Espressif Systems, “ESP32 technical reference manual,” 2020.

[7] “ESP-NOW - ESP32 - ESP-IDF Programming Guide v5.5 documentation,” Espressif.com.

[8] Mdpi.com, “Low-Power Wake-Up Radio Techniques for UAV-WSN,” Sensors, vol. 24, no. 22, p. 7170, 2024.

[9] C. Hernández-Goya, R. Aguasca-Colomo, and C. Caballero-Gil, “BLE-based secure tracking system proposal,” Wirel. Netw., vol. 30, no. 6, pp. 5759-5770, 2024.

[10] ResearchGate/USENIX/ACM DL, “High-reliability concurrent flooding (Glossy/Splash/COFlood) for NLOS bursts,” 2018.

[11] N Shylashree, S. Kumar, and H. Min, “Combined compression and encryption of linear wireless sensor network data using autoencoders,” Scientific Reports, vol. 15, no. 1, May 2025.

[12] W. Zhao and M. H. Ammar, “Message ferrying: Proactive routing in highly-partitioned wireless ad hoc networks,” Jun. 2004.

[13] S. A. Shah et al., “Collaborative UAV-WSN Systems for Civilian Applications: A Survey,” Sensors, vol. 19, no. 18, p. 4015, 2019.

[14] J. Caballero et al., “Mobile Robot SLAM-Assisted Localization of WSN Nodes,” IEEE Trans. Mobile Comput., vol. 7, no. 9, pp. 1111-1125, 2008.

[15] J. Ermis et al., “Priority-Based Data Collection for UAV-Aided Mobile Wireless Sensor Networks,” Sensors, vol. 20, no. 7, p. 2014, 2020.

[16] S. Chowdhury et al., “Data Acquisition Control for UAV-Enabled Wireless Rechargeable Sensor Networks,” Drones, vol. 7, no. 1, p. 34, 2023.

[17] A. Ramadan et al., “A Comprehensive Survey on Wireless Sensor Network Fault Diagnosis,” Information Fusion, vol. 55, pp. 60-79, 2019.

[18] N. Varghese et al., “Fault Detection and Recovery in UAV-Assisted Sensor Networks,” Computer Networks, vol. 237, p. 110013, 2024.

[19] Network visualization/dashboard tools in UAV-WSN systems. ResearchGate/Elsevier/IEEE Access, 2024.

[20] W. R. Heinzelman, A. Chandrakasan and H. Balakrishnan, “Energy-efficient communication protocol for wireless microsensor networks,” Proceedings of the 33rd Annual Hawaii International Conference on System Sciences, Maui, HI, USA, 2000, pp. 10 pp. vol.2-, doi: 10.1109/HICSS.2000.926982.

[21] O. Younis and S. Fahmy, “HEED: a hybrid, energy-efficient, distributed clustering approach for ad hoc sensor networks,” in IEEE Transactions on Mobile Computing, vol. 3, no. 4, pp. 366-379, Oct.-Dec. 2004, doi: 10.1109/TMC.2004.41.

[22] Sun, Z. Han and K. J. R. Liu, “Defense of trust management vulnerabilities in distributed networks,” in IEEE Communications Magazine, vol. 46, no. 2, pp. 112-119, February 2008, doi: 10.1109/MCOM.2008.4473092.

[23] Zahariadis, Theodore & Helen, Leligou & Panagiotis, Trakadas & Voliotis, Stamatis. (2010). Trust management in wireless sensor Networks. European Transactions on Telecommunications. 21. 386 - 395. 10.1002/ett.1413.

[24] Georgiadis, Leonidas & Neely, Michael & Tassiulas, Leandros. (2006). Resource Allocation and Cross-Layer Control in Wireless Networks. Foundations and Trends in Networking. 1. 10.1561/1300000001.

[25] Neely, Michael. (2010). Stochastic Network Optimization with Application to Communication and Queueing Systems. 10.2200/S00271ED1V01Y201006CNT007.

[26] Edwards, Sheri. (2008). Elements of Information Theory, 2nd ed., Thomas M. Cover, Joy A. Thomas. John Wiley & Sons, Inc. (2006). Information Processing & Management. 44. 400-401. 10.1016/j.ipm.2007.02.009.

[27] Deb, Kalyan. (2001). Multiobjective Optimization Using Evolutionary Algorithms. Wiley, New York.

[28] Richards, Spencer & Berkenkamp, Felix & Krause, Andreas. (2018). The Lyapunov Neural Network: Adaptive Stability Certification for Safe Learning of Dynamical Systems. 10.48550/arXiv.1808.00924.

[29] Wu, Weihua & Yang, Qinghai & Li, Bingbing & Kwak, Kyung. (2016). Adaptive Resource Allocation Algorithm of Lyapunov Optimization for Time-Varying Wireless Networks. IEEE Communications Letters. 20. 1-1. 10.1109/LCOMM.2016.2540624.

[30] Z. Fei, B. Li, S. Yang, C. Xing, H. Chen and L. Hanzo, "A Survey of Multi-Objective Optimization in Wireless Sensor Networks: Metrics, Algorithms, and Open Problems," in IEEE Communications Surveys & Tutorials, vol. 19, no. 1, pp. 550-586, Firstquarter 2017, doi: 10.1109/COMST.2016.2610578.

[31] Liu, Jianhua & Zhang, Xudong & Zhou, Haitao & Lei, Xia & Li, Huiru & Wang, Xiaofan. (2025). Lyapunov-Based Deep Deterministic Policy Gradient for Energy-Efficient Task Offloading in UAV-Assisted MEC. Drones. 9. 653.10.3390/drones9090653.

[32] Zhang, Yang & Yang, Ling & Tan, Yan. (2025). Energy-efficient adaptive routing in heterogeneous wireless sensor networks via hybrid PSO and dynamic clustering. Journal of Cloud Computing. 14. 10.1186/s13677-025-00768-3.